

PDF Mirage: Content Masking Attack Against Information-Based Online Services

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PDF Mirage: Content Masking Attack Against Information-Based Online Services

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Abstract:

We present a new class of content masking attacks against the Adobe PDF standard, causing documents to appear to humans dissimilar to the underlying content extracted by information-based services. We show three attack variants with notable impact on real-world systems. Our first attack allows academic paper writers and reviewers to collude via subverting the automatic reviewer assignment systems in current use by academic conferences including INFOCOM, which we reproduced. Our second attack renders ineffective plagiarism detection software, particularly Turnitin, targeting specific small plagiarism similarity scores to appear natural and evade detection. In our final attack, we place masked content into the indexes for Bing, Yahoo!, and DuckDuckGo which renders as information entirely different from the keywords used to locate it, enabling spam, profane, or possibly illegal content to go unnoticed by these search engines but still returned in unrelated search results. Lastly, as these systems eschew optical character recognition (OCR) for its overhead, we offer a comprehensive and lightweight alternative mitigation method.